COMSATS University Islamabad

Registrar Office, Principal Seat, Islamabad

***

Scheme of Studies of Bachelor of Science in Cyber Security for Spring 2024

1. Minimum Duration: **4 Years**

2. Minimum Semesters: **8**

2. Maximum Duration: **0 Years**

4. Maximum Semesters: **0**

	Course Work	Min No. of Courses	Min No. of Credit Hrs
1	a. General Education Courses:	12	30
2	i. Computing Core (Mandatory courses)	12	40
3	ii. Domain Core (Mandatory courses)	06	18
4	iii. Domain Elective (any Seven courses from the following list)	07	21
5	c. Interdisciplinary/Allied Courses:	05	15
6	d. Field Experience/Internship:	01	03
7	e. Capstone Project	02	06

Note:

Common policies and procedures notified vide No. CUI-Reg/Notif-1794/23/1884,dated August 25, 2023 relating to Undergraduate Degree Programs approved by the Competent Authority and amended from time to time shall be applicable.

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Application of Information and Communication Technologies (Mandatory course) :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
1	CSC101	Applications of Information and Communication Technologies	3(2, 1)		

Arts and Humanities (any one Course from the following list) :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
2	ADA111	History and Theory of Art and Culture I	2(2, 0)		
3	ARC108	Art Appreciation	2(2, 0)		
4	ARC351	Calligraphic Arts	2(2, 0)		
5	CSC210	Professional Practices	2(2, 0)		
6	HUM310	Islamic History	2(2, 0)		

B. List of Interdisciplinary Courses (Mandatory courses) :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
7		*Supporting Elective Courses	3(3, 0)		
8	HUM121	Technical and Business Writing	3(3, 0)		
9	MTH105	Multivariable Calculus	3(3, 0)		MTH104
10	MTH231	Linear Algebra	3(3, 0)		
11	MTH262	Statistics and Probability Theory	3(3, 0)		

C. Internship (Mandatory courses) :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
12	CYC395	Field Experience/Internship	3(0, 3)		

Civics and Community Engagement (Mandatory course) :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
13	HUM208	Civics and Community Engagement	2(2, 0)		

Deficiency Courses (only for Pre-Medical Students)

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
14	MTH091	Pre-Calculus I	3(3, 0)		
15	MTH092	Pre-Calculus II	3(3, 0)		

e. Capstone Project

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
16	CYC498	Final Year Project I	2(0, 2)		CSC241 ,CSC270 ,CSC291
17	CYC499	Final Year Project II	4(0, 4)		CYC498

Entrepreneurship (Mandatory course) :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
18	MGT250	Introduction to Entrepreneurship	2(2, 0)		

Expository Writing (Mandatory Course) :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
19	HUM120	Expository Writing	3(3, 0)		

Functional English (Mandatory Course) :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
20	HUM104	Functional English	3(3, 0)		

General Education Courses: :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
21	HUM101	Understanding Quran	0(0, 0)		

22	HUM117	Pakistan Studies	2(2, 0)		
23	HUM141	Seerah of Holy Prophet S.A.W.	0(0, 0)		

i. Computing Core (Mandatory courses) :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
24	CSC103	Programming Fundamentals	4(3, 1)		
25	CSC211	Data Structures	4(3, 1)		CSC103
26	CSC232	Information Security	3(2, 1)		
27	CSC241	Object Oriented Programming	4(3, 1)		CSC103
28	CSC262	Artificial Intelligence	3(2, 1)		
29	CSC270	Database Systems	4(3, 1)		
30	CSC275	Computer Networks	3(2, 1)		
31	CSC291	Software Engineering	3(3, 0)		
32	CSC301	Design and Analysis of Algorithms	3(3, 0)		CSC211
33	CSC323	Operating Systems	3(2, 1)		
34	CSC325	Computer Organization and Assembly Language	3(2, 1)		EEE240
35	EEE240	Fundamentals of Digital Logic Design	3(2, 1)		

Ideology and Constitution of Pakistan (Mandatory course) :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
36	HUM113	Ideology and Constitution of Pakistan	2(2, 0)		

ii. Domain Core (Mandatory courses) :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
37	CYC205	Introduction to Cyber Security	3(2, 1)		
38	CYC303	Digital Forensics	3(2, 1)		
39	CYC307	Information Assurance	3(3, 0)		
40	CYC365	Network Security	3(2, 1)		CSC275
41	CYC386	Secure Software Design and Development	3(2, 1)		
42	CYC390	Vulnerability Assessment and Reverse Engineering	3(2, 1)		

iii. Domain Elective (any Seven courses from the following list)

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
43	CSC303	Mobile Application Development	3(2, 1)		CSC241
44	CSC336	Web Technologies	3(2, 1)		CSC241
45	CSC337	Advanced Web Technologies	3(2, 1)		CSC336
46	CSC365	HCI and Computer Graphics	3(2, 1)		
47	CSC381	Blockchain Technology	3(3, 0)		
48	CSC412	Advanced Programming	3(2, 1)		CSC241
49	CSC417	E-Commerce and Digital Marketing	3(2, 1)		
50	CSC418	DevOps for Cloud Computing	3(2, 1)		
51	CYC309	Ethical Hacking	3(2, 1)		CSC275
52	CYC312	Cyber Threat Intelligence	3(3, 0)		
53	CYC331	Cryptography	3(3, 0)		
54	CYC332	Cryptanalysis	3(3, 0)		CYC331
55	CYC342	Web Application Security	3(2, 1)		CSC336
56	CYC350	*Topics in Cyber Security I	3(3, 0)		
57	CYC367	Security Analytics and Machine Learning	3(2, 1)		
58	CYC378	Penetration Testing	3(2, 1)		
59	CYC466	Malware Analysis	3(2, 1)		
60	CYC467	Wireless and Mobile Security	3(2, 1)		
61	CYC468	Hardware Security	3(2, 1)		
62	CYC469	Cyber Warfare	3(2, 1)		
63	CYC470	Embedded Systems	3(2, 1)		
64	CYC471	Control System Security	3(2, 1)		
65	CYC473	Embedded Systems Security	3(2, 1)		

66	CYC483	*Topics in Cyber Security II	3(3, 0)		
67	EEE341	Introduction to Computer Architecture	3(2, 1)		EEE240

Islamic Studies (Mandatory course. HUM116 Ethics is only for Non-Muslims Students) :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
68	HUM112	Islamic Studies	2(2, 0)		
69	HUM116	Ethics	2(2, 0)		

List of Supporting Elective Courses (Any one course for Supporting Elective Courses) :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
70	ECO484	Project Planning and Monitoring	3(3, 0)		
71	HUM220	Introduction to Psychology	3(3, 0)		
72	HUM221	International Relations	3(3, 0)		
73	HUM430	French	3(3, 0)		
74	HUM431	German	3(3, 0)		
75	HUM432	Arabic	3(3, 0)		
76	HUM434	Chinese	3(3, 0)		
77	MGT131	Financial Accounting	3(3, 0)		
78	MGT210	Fundamentals of Marketing	3(3, 0)		
79	MGT350	Human Resource Management	3(3, 0)		
80	MGT513	New Product Development	3(3, 0)		

Natural Sciences (any one course from the following list) :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
81	BIO100	Fundamentals of Biology	3(2, 1)		
82	BIO310	Introduction to Bioinformatics	3(2, 1)		
83	BIO321	Fundamentals of Genetics	3(2, 1)		
84	CHM100	Introduction to Chemistry	3(2, 1)		
85	PHY124	Applied Physics	3(2, 1)		

Quantitative Reasoning (any two courses from the following list)) :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
86	CSC102	Discrete Structures	3(3, 0)		
87	MTH103	Exploring Quantitative Skills	3(3, 0)		
88	MTH104	Calculus and Analytic Geometry	3(3, 0)		
89	MTH114	Tools for Quantitative Reasoning	3(3, 0)		

Social Sciences (any one course from the following list) :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
90	HUM122	Fundamentals of Psychology	2(2, 0)		
91	HUM130	Fundamentals of Sociology	2(2, 0)		
92	HUM131	Anthropology	2(2, 0)		
93	HUM209	Fundamentals of Political Science	2(2, 0)		
94	HUM222	Fundamentals of International Relations	2(2, 0)		

Note:

Note: 1. Non-Muslim students can opt for HUM114 Ethics 3(3, 0) course in lieu of Islamic Studies, if they intend to. 2. Final Year Project-I must be graded independently. 3. Deficiency Courses (Non-Credit). Students with Pre-Medical background at Intermediate level have to pass deficiency courses i.e., of MTH091 (Pre-Calculus I) and MTH092 (Pre-Calculus II)

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Note:

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